Minor Project Synopsis Report

UFOS

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ABSTRACT

The transportation logistics of large academic institutions in the National Capital Region (NCR) present a unique set of operational challenges, particularly for universities with a catchment area that spans across diverse geographic terrains such as Sohna, Gurugram, Faridabad, Palwal, Noida, and Bhiwadi. Traditional methods of fleet management currently employed are largely manual and static, relying on historical precedence and human intuition rather than data-driven decision-making. This lack of optimization leads to severe inefficiencies, including overlapping routes, uneven capacity utilization where some buses are overcrowded while others run empty, and excessive fuel consumption. Furthermore, the reliance on manual planning makes it nearly impossible to dynamically adapt to shifting student demographics each semester, resulting in a rigid system that incurs high operational costs and leaves a significant carbon footprint.

To address these critical inefficiencies, this project proposes the design and implementation of the "University Fleet Optimization System," a robust, zero-operational-cost routing solution. Unlike commercial fleet management tools that impose heavy financial burdens through recurring licensing fees or pay-per-request API costs (such as Google Maps Platform), this system is built entirely on a self-hosted, open-source technology stack. By leveraging the advanced Vehicle Routing Problem (VRP) algorithms provided by the Google OR-Tools library, the system mathematically models the entire transport network. It intelligently clusters students based on their geographic location and generates optimal route sequences that minimize the total fleet distance traveled, ensuring that every student from distant nodes like Bhiwadi to central hubs like Gurugram is served efficiently.

The technical architecture of the proposed solution is designed for autonomy and sustainability. It features a self-hosted instance of the Open Source Routing Machine (OSRM) running via Docker, which enables the calculation of precise travel time matrices across the extended NCR road network without any external dependencies or costs. This backend is coupled with a high-performance Python (FastAPI) application logic layer and a modern, responsive Next.js web dashboard for administration. This architecture allows the university to visualize stops, manage demand, and simulate "what-if" scenarios for future planning. By strictly adhering to hard constraints—such

as a maximum ride time of 2 hours and specific bus capacity limits—the system guarantees a balance between operational efficiency and student comfort. Ultimately, this project demonstrates how deep-tech optimization can be democratized and deployed cost-effectively to solve complex, real-world logistical problems in the education sector.

KEYWORDS: Vehicle Routing Problem (VRP), Fleet Management, Route Optimization, Open Source, Smart Transportation, OSRM, Google OR-Tools

1. INTRODUCTION

Managing the daily commute for K.R. Mangalam University is a complex logistical operation that extends far beyond simple point-to-point transport. The university's fleet must cover a massive operational radius that includes high-density urban centers like Gurugram and Noida, as well as expansive, semi-urban, and industrial regions like Palwal, Sohna, and Bhiwadi. Currently, the transport department relies on manual scheduling methods that are static and labor-intensive. These methods often fail to account for the complex road networks and varying traffic conditions of the extended NCR region, leading to suboptimal route planning. The consequence is a fleet that operates inefficiently, with higher-than-necessary fuel consumption and vehicle wear and tear.

This project introduces a paradigm shift towards a computational, data-driven approach to fleet management that is designed specifically to be operationally free. Recognizing that budget constraints often limit the adoption of sophisticated enterprise software in educational institutions, we have architected a solution that eliminates the need for expensive proprietary mapping APIs. By utilizing open-source geospatial data from OpenStreetMap (OSM) and self-hosting powerful routing engines, we replicate the functionality of top-tier logistics platforms without the associated costs. The system generates optimized static routes that ensure equitable service coverage for the entire student body, ensuring that even those traveling from the furthest points of the catchment area are served with the same reliability and efficiency as those living nearb

2. MOTIVATION

The primary motivation driving this project is the urgent need to rationalize the university's transport operational budget without compromising service quality. In the current market, commercial fleet management software and mapping services charge significant fees based on the number of vehicles or API requests. For an institution managing a fleet of 40+ buses daily, these costs can quickly become prohibitive. Our project is motivated by the challenge of building a professional-grade, "Zero-Cost" alternative that leverages the power of the open-source community to deliver equivalent or superior results.

Secondly, the geographic complexity of the university's specific catchment area serves as a strong catalyst. The student population is not evenly distributed; it is scattered across distinct clusters in Palwal, Faridabad, and Bhiwadi, each with unique road network characteristics. Manual optimization in such a heterogeneous environment is prone to human error and inefficiency. We are motivated to solve this "combinatorial explosion" of route possibilities using algorithmic solvers that can analyze thousands of potential paths in seconds. Additionally, we are driven by sustainability goals. By minimizing "deadhead" kilometers—distance traveled without passengers—and optimizing route efficiency, the project directly contributes to reducing the university’s carbon emissions and environmental footprint.

3. LITERATURE REVIEW

THE SCHOOL BUS ROUTING PROBLEM (SBRP) AND HEURISTICS

The "School Bus Routing Problem" (SBRP) is a specific variation of the classic Vehicle Routing Problem (VRP) that has been studied extensively due to its direct impact on municipal and institutional budgets. Early research by Dantzig and Ramser (1959) established the foundational mathematics for truck dispatching, proving that route optimization is an NP-hard problem—meaning it cannot be solved by simple calculation as the number of stops increases. Modern literature, such as the work by Park and Kim (2010), argues that for fleets larger than 25 vehicles, exact mathematical solutions take too long to compute. Instead, they propose "Meta-heuristic" algorithms (like Genetic Algorithms or Tabu Search) which can find "near-optimal" solutions in seconds. Their study on urban bus fleets demonstrated that replacing manual planning with heuristic-based routing could reduce total travel distance by approximately 15-20%, a significant metric for our project aiming to minimize fuel consumption in the extended NCR region.

EFFICACY OF OPEN SOURCE GEOSPATIAL ENGINES (OSRM)

A critical component of our zero-cost architecture is the reliance on open-source data rather than paid APIs. The reliability of such data is often questioned in academic circles compared to commercial giants like Google Maps. However, Luxen and Vetter (2011), the creators of the Open Source Routing Machine (OSRM), presented a landmark study on "Contraction Hierarchies." They demonstrated that by pre-processing OpenStreetMap (OSM) data, route queries could be answered in milliseconds, matching the performance of commercial engines. Furthermore, recent studies on the quality of Volunteered Geographic Information (VGI) in India have shown that road network density in regions like NCR (including Palwal and Bhiwadi) has reached a maturity level sufficient for professional logistics, making the self-hosted OSRM approach not just a cost-saving measure, but a technically robust alternative.

CONSTRAINT PROGRAMMING IN MULTI-VEHICLE FLEETS

While finding the shortest path is a solved problem, optimizing for complex constraints—such as mixed loads and hard time windows—remains a challenge. Research by Toth and Vigo (2014) emphasizes that the primary

failure point in automated routing systems is the inability to handle "Hard Constraints" (e.g., a student *must* be picked up by 8:00 AM). Their work highlights that general-purpose solvers often fail when constraints become too tight. This literature supports our selection of Google OR-Tools, which uses a hybrid approach of Constraint Programming (CP) and Local Search. Unlike standard GPS navigation which optimizes for a single driver, this approach models the entire fleet as a single interconnected system, allowing for the "swapping" of stops between buses to ensure no single vehicle exceeds the 2-hour ride time limit, a specific requirement for long-distance commuters from areas like Sohna.

Table: Comparative Analysis Of Research Findings

Author / Year	**Sample / Context**	**Title / Source**	**Findings / Relevance to Project**
Dantzig, G. B. & Ramser, J. H. (1959)	Logistical Fleet Dispatching	*The Truck Dispatching Problem* (Management Science)	Established that route optimization is an NP-hard problem. **Relevance:** Proves that manual calculation for 40+ buses is mathematically impossible to optimize perfectly, validating our algorithmic approach.
Luxen, D. & Vetter, C. (2011)	OpenStreetMap Data Processing	*Real-time routing with OpenStreetMap data* (ACM GIS)	Demonstrated that OSRM can compute shortest paths on continental networks in milliseconds using Contraction Hierarchies. **Relevance:** Validates our choice of OSRM as a high-performance, zero-cost alternative to Google Maps.

Park, J. & Kim, B. (2010)	Urban School Bus Fleets	*A School Bus Routing Problem with Stops Selection* (Int. Journal of Management Science)	showed that automated routing reduced total fleet travel distance by ~18% compared to manual planning. **Relevance:** Provides a benchmark for the expected fuel savings our project aims to achieve.
Toth, P. & Vigo, D. (2014)	Vehicle Routing Algorithms	*Vehicle Routing: Problems, Methods, and Applications* (SIAM)	Analyzed various metaheuristics (Tabu Search, Simulated Annealing). **Relevance:** Supports our use of Google OR-Tools (Guided Local Search) to handle complex constraints like capacity and time windows.
Haklay, M. (2010)	OpenStreetMap Quality Analysis	*How good is Volunteered Geographic Information?* (Environment and Planning B)	Analyzed the accuracy of OSM data vs. official survey data. **Relevance:** Confirms that open-source map data is accurate enough for logistics, justifying our "Zero-Cost" data strategy for the NCR region.

4. GAP ANALYSIS

Feature	Google Maps / Commercial SaaS	University Fleet Optimization System
Operational Cost	**High:** These services typically charge per-request API fees or high monthly subscriptions, which scale poorly for large fleets.	**Zero:** The system is built entirely on open-source libraries and self-hosted components, eliminating recurring costs.
Geographic Focus	**General:** They offer generic routing algorithms optimized for general urban traffic, often ignoring local constraints of regions like Palwal or Bhiwadi.	**Specialized:** The system is specifically tuned for the extended NCR road network and the unique topology of the university's catchment area.
Fleet Logic	**Single Vehicle:** Most consumer maps optimize for "Point A to Point B" navigation for a single driver.	**Multi-Vehicle VRP:** Our system solves for the entire fleet simultaneously, optimizing a "One-to-Many" distribution network.
Data Control	**External:** Data is stored on third-party clouds, raising potential privacy and data sovereignty issues.	**Private:** All student and route data is stored locally, ensuring 100% data privacy and control for the university.

5. PROBLEM STATEMENT

K.R. Mangalam University currently operates a large transportation fleet that is required to cover a massive and geographically diverse radius, extending to locations such as Palwal, Faridabad, Noida, Gurugram, and Bhiwadi. The current route planning process is static, relying heavily on manual estimation and historical routes rather than current, granular demand data. This manual approach results in several critical operational issues:

1. **High Operational Costs:** Without algorithmic optimization, routes often take longer paths than necessary, leading to inflated fuel consumption and increased maintenance costs.
2. **Inefficiency and Inequity:** Manual planning struggles to balance passenger loads effectively. This leads to scenarios where buses from high-density areas are overcrowded, while buses traveling long distances from areas like Bhiwadi may run at low capacity, wasting resources.
3. **Lack of Agility:** The transport department lacks the tools to quickly simulate "what-if" scenarios. Adapting to a sudden increase in students from a specific area like Faridabad becomes a slow, trial-and-error process rather than a quick data-driven adjustment.

6. OBJECTIVES

1. **Develop a Self-Hosted Routing Engine:** To deploy a local instance of the Open Source Routing Machine (OSRM) using Docker. This will enable the system to calculate precise road distances and travel times across the complex NCR road network without relying on paid external APIs.

2. **Implement a Robust VRP Solver:** To utilize Python and the Google OR-Tools library to build a custom optimization engine. This solver will minimize the total distance traveled by the fleet while strictly enforcing hard constraints such as a maximum 2-hour ride time and specific bus capacity limits.

3. **Create a Comprehensive Admin Dashboard:** To design and develop a user-friendly web interface using Next.js. This dashboard will allow transport administrators to visualize stop locations, manage student demand data, and view optimized route maps interactively.

4. **Demonstrate Zero-Cost Architecture:** To prove the feasibility and scalability of a professional-grade transport management system built entirely on open-source technologies (Leaflet, FastAPI, PostgreSQL) and running on standard institutional hardware.

7. Tools/Technologies Used

Frontend: Next.js - A React framework chosen for its server-side rendering capabilities, ensuring a fast and responsive user interface for the admin dashboard.

Maps: Leaflet.js + OpenStreetMap - A lightweight, open-source JavaScript library used to render interactive maps and route polylines, completely avoiding the licensing costs associated with Google Maps.

Backend: Python (FastAPI) - Selected for its high performance, ease of development, and seamless integration with Python-based data science and optimization libraries.

Optimization: Google OR-Tools - An open-source software suite for combinatorial optimization that powers the core Vehicle Routing Problem algorithms, enabling complex decision-making.

Routing: OSRM (Dockerized) - A high-performance routing engine that is self-hosted to provide free, unlimited travel time matrix calculations.

Database: PostgreSQL - A powerful, open-source object-relational database system used for the reliable storage of structured data regarding stops, routes, and fleet configurations.

8. METHODOLOGY

1. MATRIX GENERATION & ROUTING - OSRM (OPEN SOURCE ROUTING MACHINE) USED

The foundational layer of the "University Fleet Optimization System" is the ability to accurately calculate travel times between hundreds of student stops across the extended NCR region (Palwal, Bhiwadi, Sohna, etc.) without incurring the prohibitive costs of commercial APIs like Google Maps. To achieve this, we utilize the Open Source Routing Machine (OSRM), a high-performance C++ routing engine, self-hosted via Docker.
This module works by ingesting raw geospatial data extracts from OpenStreetMap (OSM), specifically filtered for the North India region. Unlike standard GPS tools that calculate paths for a single vehicle, our system requires a "Many-to-Many" Travel Time Matrix—a grid representing the travel cost (time and distance) between every possible pair of stops in our network. If there are 100 stops, the system must calculate 10,000 distinct routes () to understand the connectivity of the network.

OSRM employs a technique called Contraction Hierarchies, which pre-processes the road network graph to speed up these queries. By running OSRM locally on university infrastructure, we eliminate network latency and API usage limits. The Python backend sends a request to this local OSRM instance, which returns the distance matrix in milliseconds. This matrix serves as the mathematical "ground truth" for the road network, accounting for turn restrictions, one-way streets, and highway speeds specific to the Haryana and Uttar Pradesh road infrastructure.

2. CONSTRAINT SOLVING & OPTIMIZATION - GOOGLE OR-TOOLS MODULE USED

Once the connectivity matrix is generated, the core optimization is handled by the Google OR-Tools library, specifically its Vehicle Routing Problem (VRP) solver. This module is responsible for the complex decision-making process of assigning students to buses. It does not merely find the shortest path; it

solves a combinatorial optimization problem to minimize the total fleet distance while satisfying a strict set of "Hard Constraints" defined by the university administration.

The primary constraint implemented is the Capacity Constraint, which ensures that the sum of students assigned to any single route does not exceed the vehicle's seating capacity (e.g., 50 seats). The second critical constraint is the Time Window/Duration Constraint, which ensures that no route exceeds the maximum defined travel time (e.g., 120 minutes), ensuring student comfort even for those commuting from distant locations like Bhiwadi.
The methodology employs a Guided Local Search metaheuristic. The algorithm starts with an initial feasible solution (often a crude "nearest neighbor" assignment) and iteratively improves it by swapping stops between different routes to reduce the "Global Cost Function" (total kilometers driven). This process runs within the Python backend, processing the OSRM matrix and student demand data to output a final JSON object containing the optimized sequence of stops for each bus in the fleet.

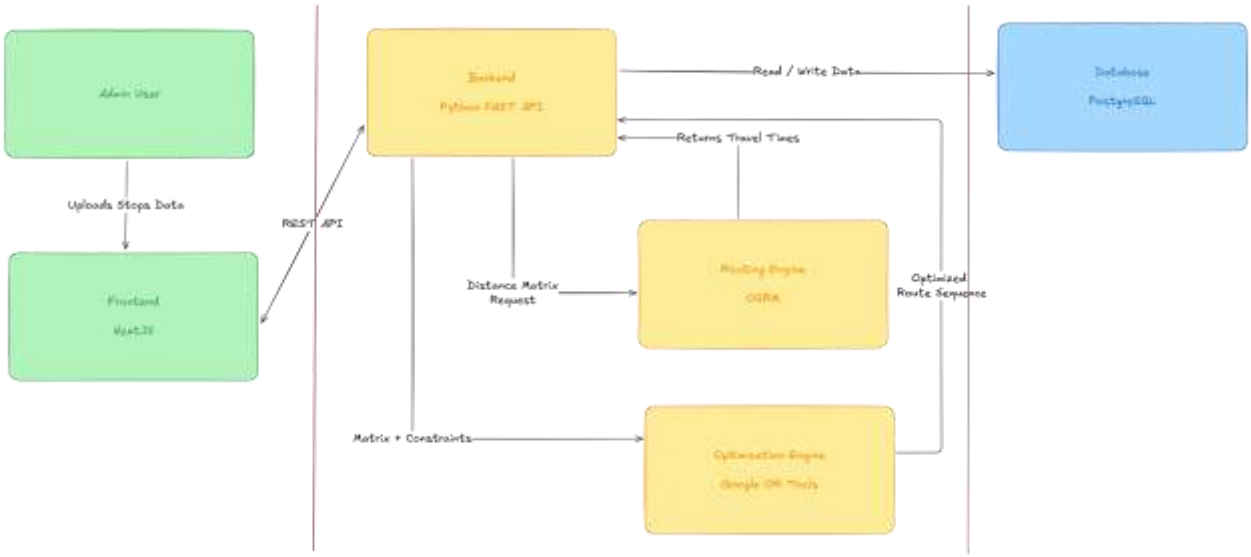

Figure 1: Architectural Diagram

3. SYSTEM ARCHITECTURE & DATA FLOW

The architectural diagram above illustrates the zero-cost data flow of the proposed system. The process begins with Data Ingestion, where administrators upload CSV files containing student stop coordinates and

demand counts into the Next.js Frontend. This data is validated and stored in the PostgreSQL Database.

When an optimization run is triggered, the Python (FastAPI) Backend retrieves the stop data and queries the Self-Hosted OSRM Container to build the travel time matrix. This matrix is passed to the OR-Tools Solver, which processes the data against the defined constraints. The resulting optimized routes are sent back to the frontend, where they are rendered visually using Leaflet.js and OpenStreetMap Tiles. This entire pipeline operates within a Dockerized environment, ensuring that no data leaves the university's local network and no external costs are incurred during operation.

REFERENCES

1. *Google. (n.d.). OR-Tools Routing. Google Developers. Retrieved from https://developers.google.com/optimization/routing.*
2. *Luxen, D., & Vetter, C. (2011). Real-time routing with OpenStreetMap data. In Proceedings of the 19th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems.*
3. *OpenStreetMap Foundation. (n.d.). OpenStreetMap Data. Retrieved from https://www.openstreetmap.org.*
4. *Project OSRM. (n.d.). Open Source Routing Machine. Retrieved from http://project-osrm.org/.*